The cost of a mark: Newton's vanishing sagitta and a floor of order $\hbar$

Abstract. Newton reads a force off a trajectory by letting the sagitta and the area between the inertial tangent and the curve go to zero, keeping their ratio to the time. In the quantum instrument model with a supplied $\hbar > 0$, a preparation-independent test of the two motions obeys a disturbance bound. For every protocol of marks, made by any instruments at any times, a test that decides at error probability ϵ between free motion and a constant force F over a duration τ, and must do so whatever the body's initial state, the state under the force being allowed to differ by an unknown displacement, requires

$$\frac{s}{8}\sum_j \Delta(\hat{D}_j) + \frac{J}{2}\sum_j \Delta(\hat{X}_j) \;\geq\; \hbar \arcsin(1-2\epsilon), \qquad s = \frac{F\tau^2}{2m}, \quad J = F\tau,$$

where $\Delta(\hat{D}_j)$ and $\Delta(\hat{X}_j)$ are the spreads of the momentum and position disturbances of mark j, with suprema over the conditional and translated hybrid states used in the proof. Newton's sagitta pairs with momentum disturbance and his impulse with position disturbance; these are operator spreads, not a theorem about the conditional uncertainty of a classically realized kick. A protocol that knows the body's preparation escapes the floor, and pays instead in the body's spread along the way. For non-adaptive marks with uncorrelated Gaussian probes, read by tests invariant under the unknown initial position and velocity, the floor takes the sharp form $\tau\Delta E = F^2\tau^3/2m \geq 24 z_{1-\epsilon}^2 \hbar$, about $65\hbar$ at five per cent error, because a mark's error and the impulse it delivers are canonically conjugate; a window of marks shorter than $\tau_* = (48 z_{1-\epsilon}^2 m\hbar/F^2)^{1/3}$, used by itself, cannot tell the force from free motion. Correlated and grid-state probes lower that form, and the disturbance bound holds for them. The comparison also carries a quantity that no preparation of the body can move, a phase: Newton's polygon inscribed in the parabola agrees with it classically at every vertex and differs quantum mechanically by $F^2\sum_j \tau_j^3/(24m\hbar)$, the parabolic segments of the chords times $F/2v\hbar$, so that each round of Archimedes' exhaustion removes a phase. On the historical side, Newton's *Opticks* contains a measured least length, the 1/89000th part of an inch, corpuscles that carry an impulse, and a disposition whose period times the corpuscle's momentum is invariant under refraction. A proposition joining these source quantities to an irreducible recording cost would be an additional physical premise. The recorded periods are not transverse standard deviations, and deterministic transport with sufficiently rich remote readout supplies a classical countermodel to that premise. The modern disturbance bounds assume quantum states, instruments and a positive canonical commutator scale; they do not derive that scale from Newton's axioms. On fixed collision-free intervals, the Kepler comparison recovers classical centroids from Coulomb evolution. The coherent-record argument requires coherence between launch points and a specified record topology; relativistic action thresholds require a selected point-source domain. These premises leave the independent necessity and universal calibration of the scale open.

Working manuscript, revised 2026-10-29. The canonical proofs are marking models, quantum resource bounds and polygon phases; the independently useful manuscript retains their historical argument and its own source passages. The recorded model/coordinator checks were internal, not independent peer review. Gaussianity, preparation invariance, hybrid-state resource bounds and the Coulomb domain qualifications are stated where used. Historical interpretations remain readings, with submission obligations in §11.

1. The question

Galileo's comparison is an inertial horizontal line against a falling parabola. With mass m, transverse force F, horizontal speed v and duration τ, the two paths are $q_{\mathrm{I}}(t) = (vt, 0)$ and $q_{\mathrm{F}}(t) = (vt, Ft^2/2m)$, and the region between them has

$$A = \frac{vF\tau^3}{6m}, \qquad s(\tau) = \frac{F\tau^2}{2m}, \qquad \Delta E = \frac{F^2\tau^2}{2m}, \qquad \frac{3F}{v}A = \tau\Delta E. \tag{1}$$

The sagitta s, the area A and the product $\tau\Delta E$ all scale as powers of τ and vanish together. Newton's Lemma X makes the first displacement quadratic in the time. Lemma XI (Corollary 3 of the first edition) makes the area between tangent and arc two thirds of the tangent triangle and the segment between chord and arc one third, "ex natura Parabolæ", both cubic in the time; the area A is the first of these. It is also Democritus's third: the sagitta grows as the square of the time as a cone's section grows as the square of the height, so $A = \frac{1}{3}v\tau s$, one third of the rectangle, as the cone is one third of its cylinder (§9). Proposition VI reads the force off the limit $2ms/\tau^2$; and Proposition I builds an orbit as a polygon struck by impulses at its vertices, the number of vertices "augmented in infinitum". The construction is designed so that the quantities in (1) disappear and their ratios survive.

Our question is what stops that refinement when the trajectory has to be recorded rather than contemplated, and whether the stopping point can be reached from materials Newton possessed.

The answer has five parts. The geometry alone supplies no floor (§2). Recording supplies one: in a Gaussian model of marks it is set by a single number κ with the dimensions of action, sharply, in the combination $F^2\tau^3/m$ (§3), and quantum kinematics fixes $\kappa \geq \hbar/2$ because a mark's error and its recoil are conjugate (§4). Within that quantum model, a preparation-independent test with probes and instruments of every kind obeys a bound on disturbance, paired with Newton's sagitta and impulse (§5), and a phase that no preparation of the body can move carries the same quantity through Newton's inscribed polygon (§6). Newton's optics supplies a measured period and a corpuscular hypothesis that motivate a comparison with probe scales. Joining them to an irreducible recording cost or a quantum phase rule requires additional physical premises (§§7–8). Newton also drafted classical scholia founded on the ancients, and the ancient arguments that bear on his own method are the ones he passed over; stated as premises about records they generate a hierarchy whose first rung is Zeno's arrow and whose second is the comparison above (§9).

2. The geometry has no floor

Proposition 1. For every partition $0 = t_0 < \cdots < t_n = \tau$ with steps τ_j and mesh $|\pi|$, the vertex impulse $F\tau_j$, the step sagitta $F\tau_j^2/2m$ and the step area $vF\tau_j^3/6m$ tend to zero with the mesh; the areas sum to at most $vF\tau|\pi|^2/6m$; and $2m\,s(\tau_j)/\tau_j^2 = F$ exactly at every step.

The proof is immediate from (1). Berkeley's objection, that evanescent increments are "neither finite Quantities nor Quantities infinitely small, nor yet nothing", is met inside the geometry by the limit ratio, which is what Newton's closing Scholium to Book I Section I provides. A floor must therefore come from the act of marking the curve, and the reading of "inserting a point" used here is physical: a point of the trajectory is inserted by making a mark on the body at a time.

3. Marks, and the floor they impose

Definitions. A **mark** at time t_j leaves a datum about the transverse coordinate. Its **resolution** δ_j is the spread of the resulting position estimate's error; its **recoil** Δ_j is the spread of the transverse impulse it delivers to the body. A protocol is a finite set of marks; it **decides** the comparison at error probability ϵ when some test on the full record does so with equal priors. A model satisfies the **mark trade-off** with constant κ when every available mark has $\delta\Delta \geq \kappa$.

The statistical model. Mark j returns $R_j = y(t_j) + \xi_j$ with ξ_j centred Gaussian of standard deviation δ_j, and delivers an impulse ι_j, centred Gaussian of standard deviation Δ_j, independently, so that

$$y(t) = y_0 + v_0 t + \theta P(t) + \frac{1}{m}\sum_j \iota_j (t - t_j)_+, \qquad P(t) = \frac{F t^2}{2m},$$

with $\theta = 0$ under the inertial hypothesis and $\theta = 1$ under the falling one, and y_0, v_0 unknown. The marks are fixed in advance, and tests are invariant under the two unknowns, which is minimax for this location family by the Hunt–Stein theorem (Lehmann and Romano, *Testing Statistical Hypotheses*); the statistic is $u^{\mathsf T} R$ with $\sum_i u_i = 0$ and $\sum_i u_i t_i = 0$, and the optimal invariant test has error probability $\Phi(-d/2)$ with $d^2 = \sup (u^{\mathsf T} P)^2 / (u^{\mathsf T} \Sigma u)$ over such u.

Theorem 2. For every non-adaptive protocol satisfying the mark trade-off with constant κ, read by tests invariant under the unknown initial position and velocity,

$$d^2 \;\le\; \frac{F^2 \tau^3}{24\, m \kappa},$$

independently of the number of marks and of their times, and the constant is sharp. Hence deciding at error probability ϵ requires

$$F^2 \tau^3 \;\ge\; 96\, z_{1-\epsilon}^2\, m\kappa, \qquad \tau \Delta E \;\ge\; 48\, z_{1-\epsilon}^2\, \kappa, \qquad A \;\ge\; \frac{16\, z_{1-\epsilon}^2\, v \kappa}{F},$$

with $z_{1-\epsilon} = \Phi^{-1}(1-\epsilon)$.

Proof. The noise enters R in two ways: ξ_j affects R_j alone, and ι_j affects R_i for $i > j$ with coefficient $(t_i - t_j)/m$. Hence

$$u^{\mathsf T} \Sigma u = \sum_j \delta_j^2 u_j^2 + \sum_j \frac{\Delta_j^2}{m^2} S_j^2, \qquad S_j = \sum_{i > j} u_i (t_i - t_j),$$

and the arithmetic–geometric mean inequality applied termwise gives $u^{\mathsf T} \Sigma u \ge \frac{2\kappa}{m} \sum_j |u_j| |S_j|$.

Introduce $N(w) = \sum_{i : t_i > w} u_i$ and $T(w) = \int_w^\tau N$. Since $\sum_i u_i = 0$, N vanishes for $w < 0$ and for $w \ge t_k$; T is continuous, piecewise linear, zero on $[\tau, \infty)$, and zero on $(-\infty, 0]$ as well, since $T(0) = \int_0^\tau N = \sum_i u_i t_i = 0$ by the invariance under v_0; it has $T' = -N$ jumping by u_j at t_j and $T(t_j) = S_j$. Two identities follow. Fubini and one integration by parts, using $P'(0) = 0$ and $T(\tau) = 0$, give

$$u^{\mathsf T} P = \int_0^\tau P' N = -\int_0^\tau P' T' = \int_0^\tau P'' T = \frac{F}{m} \int_0^\tau T;$$

and since T' has bounded variation with jumps u_j and no other variation, while $T T'$ vanishes at both ends,

$$\sum_j u_j T(t_j) = \int_{\mathbb R} T \, dT' = -\int_{\mathbb R} T'^2 = -E, \qquad E = \int_0^\tau T'^2.$$

Therefore $\sum_j |u_j| |S_j| \ge E$ and $u^{\mathsf T} \Sigma u \ge 2\kappa E / m$. Since T vanishes at both ends of $[0, \tau]$, the Poincaré inequality $(\int_0^\tau T)^2 \le \frac{\tau^3}{12} \int_0^\tau T'^2$ applies, with equality for $T \propto w(\tau - w)$ (the Euler–Lagrange equation is $T'' = \text{const}$), and

$$\frac{(u^{\mathsf T} P)^2}{u^{\mathsf T} \Sigma u} \le \frac{(F/m)^2 \frac{\tau^3}{12} E}{2\kappa E / m} = \frac{F^2 \tau^3}{24 m \kappa}.$$

Finally $\Phi(-d/2) \le \epsilon$ requires $d \ge 2 z_{1-\epsilon}$. For sharpness, take marks dense in $[0, \tau]$ with weights making N approximate $2w - \tau$, so that T approximates $w(\tau - w)$; the interior weights are negative where T is positive, so every $u_j S_j$ has one sign and $\sum_j |u_j| |S_j| = E$; balance each mark, $\delta_j |u_j| = \Delta_j |S_j| / m$, so the arithmetic–geometric step is an equality; and make the end marks sharp, which costs nothing since $S = T = 0$ there. Every inequality then becomes an equality in the limit. □

Three features of the proof carry the claim that the bound holds for every protocol of the model. The certificate u ranges over all invariant tests, so no monitoring scheme is privileged. The function T encodes the whole protocol in one object, and the bound depends on it only through the Dirichlet energy E, which cancels. And the exponent 3 on τ comes from the Poincaré constant of the interval, $\tau^3/12$, which is where the combination in (1) originates. The sharp constant is approached, never attained, by dense protocols whose end marks are sharp, with a recoil that no invariant statistic sees.

The floor rests on invariance. The body's initial operators enter every reading as $\hat{y}_0 + \hat{p}_0 t_j/m$ and cancel from every invariant statistic, so the body's state plays no part in Theorem 2. A protocol that knows the preparation escapes the floor even with Gaussian states: a contractive Gaussian state focused at time τ and one sharp final mark decide at every force, which is Yuen's protocol (PRL **51**, 719, 1983). The boundary is knowledge of the preparation, and §10 prices the preparation route by the body's spread along the way.

Corollary 3 (insertion). Marks inside a window of duration τ', used by themselves, record the force only if $F^2\tau'^3 \geq 96 z_{1-\epsilon}^2 m\kappa$. So a window shorter than

$$\tau_* = \left(\frac{96\, z_{1-\epsilon}^2\, m\kappa}{F^2} \right)^{1/3},$$

used by itself, is consistent with free motion at the stated confidence. Refinement continues below τ_*, and the sharp protocol of Theorem 2 needs marks far denser than that; what stops is local exhibition of the force. Newton's polygon may be refined past τ/τ_* vertices, and its impulses remain in the geometry, though no record confined to one such window exhibits them.

A worst-case counterpart. If the marks instead confine position and impulse to intervals with certainty, the same three steps run with a separating hyperplane in place of the optimal test direction and give $F^2\tau^3 > 9m\kappa$ for every protocol, with two marks sufficing above $36m\kappa$. There the initial velocity is bounded by the preparation's recoil width instead of being free, so T need not vanish at 0 and the Cauchy–Schwarz supplies its non-sharp lower constant. This is the full independent adversarial-box model, and the upper construction requires the specified balanced widths to be available. It assigns no quantum support-width value to κ: Paley–Wiener concerns position/momentum of a state and does not alone prove a universal statement about arbitrary-instrument recoil. The canonical marking treatment gives the exact hypotheses and corrected area conversions.

4. The cost of a mark is the probe's uncertainty product

Theorem 4. Let a mark be realized by the impulsive coupling of the body's position to a probe, $U = \exp(-\frac{i}{\hbar}\lambda\hat{y}\hat{P}_A)$ with $[\hat{Q}_A, \hat{P}_A] = i\hbar$, the pointer $\hat{Q}_A$ being read afterwards. Then

$$U^\dagger \hat{Q}_A U = \hat{Q}_A + \lambda\hat{y}, \qquad U^\dagger \hat{p} U = \hat{p} - \lambda\hat{P}_A,$$

so the error operator $\hat{N} = \hat{Q}_A/\lambda$ and the delivered impulse $\hat{D} = -\lambda\hat{P}_A$ satisfy $[\hat{N}, \hat{D}] = -i\hbar$ and

$$\kappa = \delta\Delta = \Delta\hat{N} \cdot \Delta\hat{D} = \Delta\hat{Q}_A \cdot \Delta\hat{P}_A \;\geq\; \frac{\hbar}{2}$$

for every probe state, pure or mixed.

Proof. With $A = \frac{i}{\hbar}\lambda\hat{y}\hat{P}_A$ the commutators $[A, \hat{Q}_A] = \lambda\hat{y}$ and $[A, \hat{p}] = -\lambda\hat{P}_A$ are central, so the Baker–Campbell–Hausdorff series terminates and gives the two displayed relations, while $\hat{y}$ and $\hat{P}_A$ are unchanged. Estimating the body's position by $\hat{Q}_A/\lambda$ leaves the error $\hat{N} = \hat{Q}_A/\lambda$, and $[\hat{N}, \hat{D}] = -[\hat{Q}_A, \hat{P}_A] = -i\hbar$. Robertson's inequality closes it. □

The coupling strength cancels, so the cost is a property of the mark itself, whatever the strength of the coupling. The record and the recoil are themselves incompatible, since $[\hat{Q}_A +$

$\lambda\hat{y}, -\lambda\hat{P}_A] = -i\hbar\lambda$, which is why no reading of the pointer determines the impulse. A deterministic probe, whose later position records fix its momentum, has $\kappa = 0$ and no floor; this is the exact point at which classical record models differ.

For Gaussian probe states whose position and momentum are uncorrelated, a fresh probe for each mark, and the linear dynamics of the comparison, the Wigner function is a probability density evolving classically, so the statistical model of §3 reproduces the quantum experiment exactly for every invariant statistic; the body's own state cancels from those and need not be Gaussian. With Theorem 4,

$$\tau\Delta E \;\geq\; 24\, z_{1-\epsilon}^2\, \hbar, \qquad \tau_* = \left(\frac{48\, z_{1-\epsilon}^2\, m\hbar}{F^2}\right)^{1/3}, \tag{2}$$

about $65\hbar$ at $\epsilon = 0.05$ and $24\hbar$ at $\epsilon = 0.16$.

Correlated marks. A probe whose position and momentum are correlated delivers an error and a recoil with correlation $\rho = -\operatorname{Cov}(\hat{Q}_A, \hat{P}_A)/(\Delta\hat{Q}_A\Delta\hat{P}_A)$, and the Robertson–Schrödinger inequality gives $\delta^2\Delta^2(1-\rho^2) \geq \hbar^2/4$. The cross terms enter the proof of Theorem 2 as $2c_j u_j S_j/m$, and $ax^2+2cxy+by^2 \geq 2(\sqrt{ab}-|c|)|x||y|$ replaces the arithmetic–geometric step, so every protocol whose marks have $|\rho_j| \leq \rho$ needs

$$\tau\Delta E \;\geq\; 24\, z_{1-\epsilon}^2\, \hbar \sqrt{\frac{1-\rho}{1+\rho}}, \tag{3}$$

sharp by the dense construction of Theorem 2 with every mark at correlation ρ of the helpful sign. Three marks at $0, \tau/2, \tau$ with the test $R_1 - 2R_2 + R_3$ come within a factor $4/3$, since the outer marks' recoils are invisible to the test and the middle mark's noise $-2\xi_2 + (\tau/2m)\iota_2$ is one quadrature of its probe: they decide once $\tau\Delta E \geq 32z^2\hbar\sqrt{(1-\rho)/(1+\rho)}$, at every force as $\rho \to 1$. This is the mechanism of Yuen's contractive states (PRL **51**, 719, 1983), which correlate the body's position and momentum, applied to the meter, as in the back-action-evading measurements of Braginsky and Khalili. The proofs are in the mark-cost note, Theorem C and Proposition D.

5. Every probe and every instrument: the record costs disturbance

All general-instrument inequalities in this section assume the common operator domains, differentiable displacement paths and finite variance envelopes specified in the canonical proof. Those envelopes cover translated hybrid conditional states as well as the physical endpoint experiments. The unknown-preparation theorem uses a test valid for every pair of initial states, a stronger requirement than the same known preparation under both hypotheses.

The floors of §§3–4 assume Gaussian probes, and the assumption is needed. A grid state of the Gottesman–Kitaev–Preskill kind with a real wavefunction has no position–momentum correlation, yet as the middle probe of the three-mark protocol it turns the statistic into a comb of spacing $\sqrt{2\pi\hbar\tau/m}$, and along the standard family of approximate grid states the teeth narrow, so the error tends to zero at any force whose signal is not a multiple of that spacing, at the price of a recoil that diverges. Grid states read both quadratures of a small displacement at once, the principle of the displacement sensor of Duivenvoorden, Terhal and Weigand (PRA **95**, 012305, 2017). What holds for every probe state is a bound on the recoil, and for every instrument a bound on disturbance.

Theorem 5. For every protocol of momentum-transfer marks with their pointers read and probes in arbitrary states, deciding at error ϵ whatever the body's initial state under each hypothesis requires $\arcsin(1-2\epsilon) \leq \hbar^{-1} \min_{a,b} \sum_j \Delta_j |P(t_j) - a - bt_j|$, with Δ_j the recoil spread of mark j. For the constant force,

$$s\sum_j \Delta_j \;\geq\; 8\hbar \arcsin(1-2\epsilon), \qquad s = \frac{F\tau^2}{2m}, \tag{4}$$

which at certain decision reads $s\sum_j \Delta_j \geq 4\pi\hbar = 2h$.

The proof moves the signal into the pointers: translating the body's initial state along the best straight line $a + bt$ turns the force into a translation of each pointer by $\lambda_j c_j$, with $c_j = P(t_j) - a - bt_j$ the line's residual at the mark, and a pointer registers a translation only through the spread of its momentum, which is the impulse the body receives. Mandelstam–Tamm and the triangle inequality over the probes finish it, and running the same chain with the Bures angle in place of total variation gives the constant $\arcsin(1-2\epsilon)$, the sharp single-shot one (path-length note, Theorem P). With sharp end marks the minimum over lines is the line through them, and the three-mark protocol with the best Gaussian middle probe, whose statistic then has spread exactly $\hbar/\Delta_2$, comes within a factor $z_{1-\epsilon}/\arcsin(1-2\epsilon)$ of that bound at every recoil, about 1.47 at five per cent error; uncorrelated probes come within $\sqrt{2}$ times that. The constant-force form (4) charges the end marks as well, and their unbounded recoil makes its left side infinite. The proofs, the grid construction and the tightness computation are in the recoil note. It is the recoil clause of M3, and Theorem 6 completes it.

The recoil bound reaches well beyond von Neumann's coupling. For any coupling and any pointer, the pointer commutes with the body's momentum after the mark, which gives $[\hat{N}, \hat{D}] = -i\hbar - [\hat{y}, \hat{D}] - [\hat{N}, \hat{p}]$. So every mark whose error and impulse are independent of the body has conjugate error and impulse, which is Ozawa's class of measurements with independent intervention (PRA **67**, 042105, 2003; Ann. Phys. **311**, 350, 2004); if it also leaves the position alone it is a von Neumann mark on some canonical pair of the apparatus, a dilation form of Davies's translation-covariant instruments (1976); and Theorem 5 holds for all such marks, position-displacing ones included (additive-noise note). When the noise depends on the body, the offset between the hypotheses still has to be carried through each mark, and the cost of carrying it is fixed by what the mark does to the body.

Theorem 6 (every instrument). Let mark j be any unitary coupling of the body to an apparatus followed by reading a pointer, with momentum and position disturbances $\hat{D}_j = U_j^\dagger \hat{p} U_j - \hat{p}$ and $\hat{X}_j = U_j^\dagger \hat{y} U_j - \hat{y}$, operators on body and apparatus that may depend on both. Deciding at error ϵ whatever the body's initial state under each hypothesis requires, for every line $a+bt$, $\arcsin(1-2\epsilon) \leq \hbar^{-1}\sum_j \sup \Delta(c_j\hat{D}_j - mc'_j\hat{X}_j)$ with $c_j = P(t_j) - a - bt_j$, the supremum over the states that enter the mark in the interpolating processes of the proof; for the constant force,

$$\frac{s}{8}\sum_j \Delta(\hat{D}_j) + \frac{J}{2}\sum_j \Delta(\hat{X}_j) \;\geq\; \hbar \arcsin(1-2\epsilon), \qquad J = F\tau. \tag{5}$$

The constants $1/8$ and $1/2$ are the price of an initial state that may differ between the hypotheses: a test allowed to assume the same initial state under both faces the coefficients s and J.

The proof is a hybrid argument that carries the offset (c, mc') between the hypotheses through the marks one at a time. Passing mark j applies the unitary $T^\dagger U_j T U_j^\dagger$, with T the translation by the offset; its generator, conjugated back to the state entering the mark, is $(c_j\hat{D}_j - mc'_j\hat{X}_j)/\hbar$, and Mandelstam–Tamm bounds its effect by that operator's spread. The line of best uniform approximation gives $|c_j| \leq s/8$ and $m|c'_j| \leq J/2$ (disturbance note). Von Neumann marks give Theorem 5, and marks that read momentum pay in position jumps.

Equation (5) involves the disturbances alone, so the error–disturbance relations, Ozawa's included (PRA **67**, 042105, 2003), are unused. Newton's two quantities appear with the partners they have in the aperture bound of §10: the sagitta with momentum, the impulse with position.

6. The part no preparation can move

The floors of §§3–5 are paid in spreads, of resolution, recoil or disturbance, and a protocol can move a spread from one quantity to another, as squeezing and grid states do. The comparison also carries a quantity with no spread at all. Newton's polygon of Proposition I inscribed in the parabola, with each step's impulse $F\tau_j$ split equally between its ends, moves uniformly along each chord and has the parabola's position and velocity at every vertex, so classically the two are the same state there.

Theorem 7. For every partition of $[0,\tau]$ into steps τ_j, the two evolutions satisfy $W_{\mathrm{poly}} = e^{i\theta_N} W_F$ with

$$\theta_N = \frac{F^2}{24m\hbar}\sum_j \tau_j^3 = \frac{F}{2v\hbar}\sum_j S_j,$$

$S_j = vF\tau_j^3/12m$ being the parabolic segment cut off by the j-th chord. Inserting a vertex at fraction λ of a step lowers θ_N by $F^2\tau_j^3\lambda(1-\lambda)/(8m\hbar)$, the inscribed triangle over $\hbar$, so midpoint halving removes it in the proportions of Archimedes' exhaustion. The same increments composed in reverse order differ by the phase $\Phi_N = F^2(\tau^3 - \sum_j \tau_j^3)/(6m\hbar)$, and $\Phi_N + 4\theta_N = \tau\Delta E/3\hbar$ for every partition.

The proof uses the Weyl relations alone: an impulse J at time t is a phase-space displacement, two histories with equal endpoints differ by the enclosed phase-plane area over $\hbar$ times the identity, and the per-step lobe is $F^2\tau_j^3/24m$ (the polygon-lift note has it in full). Since $W_{\mathrm{poly}} W_F^\dagger$ is a multiple of the identity, no preparation of the body, squeezed or mixed, changes θ_N. A control qubit that selects the history reads it at single-shot error $\frac{1}{2}(1 - |\sin(\theta_N/2)|)$, so in one controlled pass a step is told from its chord at error ϵ only if $\tau_j \geq (48m\hbar \arcsin(1-2\epsilon)/F^2)^{1/3}$, for every state of the body; k passes accumulate $k\theta_N$ and lower that threshold by $k^{1/3}$, so the readout's resource is the number of passes. The single chord lens and the closed loop of the earlier notes are instances. In numerical analysis the inscribed polygon is the kick–drift–kick (Strang) splitting of the propagator, exact for a linear potential up to this c-number; what is new is its reading in Lemma XI's areas and Archimedes' segments.

Archimedes' exhaustion, read through $\hbar$. Archimedes squares the parabola by inscribing, in the segment cut off by a chord, the triangle whose apex is "the vertex of the segment", where the tangent is parallel to the chord; the two segments left over receive their own triangles, each round totalling a quarter of the one before, and his Proposition 23 sums the series, $A + B + C + \cdots + Z + \frac{1}{3}Z = \frac{4}{3}A$, so that the segment is four thirds of its first triangle (Proposition 24). On the Galileo parabola traced in time the vertex of a step's segment is its midpoint in time, where the velocity equals the chord's slope, so his construction is Newton's refinement by halving, and the triangles he adds are those Theorem 7 removes. With $T_j = \frac{3}{4}S_j$ the first triangle,

$$\theta_{\mathrm{step}} = \frac{F}{2v\hbar}\cdot\frac{4}{3}T_j = \frac{F}{2v\hbar}\Big(T_j + \frac{T_j}{4} + \frac{T_j}{16} + \cdots\Big),$$

and Proposition 23 becomes the statement that the phases removed by all the rounds add up to the phase between polygon and parabola. Exhaustion shows classically that the leftover segments, 4^{-k} of the whole after k rounds, fall below any given area. With $\hbar$ they are phases, and two scales appear: each segment's phase falls below a radian once the steps are shorter than $\tau_\hbar = (24m\hbar/F^2)^{1/3}$, and their total once 4^k exceeds $(\tau/\tau_\hbar)^3$. The method began with Democritus's cone, which Eudoxus proved to be a third of its cylinder (§9), and the Galileo area is the same third, $A = \frac{1}{3}v\tau s$. The two figures differ in what $\hbar$ does to them: the cone is static, its sections can be marked without disturbing it, and Theorem 9(iii) finds no floor; the parabola is traced by a motion, and its exhaustion carries phases.

Theorem 8 (every force law). For a force history f on an interval of duration τ, let $\mathcal{K}_\tau[f] = \int \frac{1}{2}m(\dot{y}_f - \dot{y}_{\mathrm{chord}})^2 dt = \frac{1}{2m}\iint G_\tau f f$ be the kinetic action of the motion relative to its

chord, $G_\tau(s,u) = \min(s,u)(\tau - \max(s,u))/\tau$. Newton's polygon, with each step's impulse split between its ends so as to match the step's displacement, differs from the motion by the phase $\sum_j \mathcal{K}_{\tau_j}[f]/\hbar$, which refinement never raises; and every protocol of uncorrelated Gaussian marks has $d^2 \le \mathcal{K}_\tau[f]/\kappa$, sharp for f of one sign. At $\kappa = \hbar/2$ the best recorded deflection is twice the phase between the motion and its chord. The proofs are in §6 of the polygon-lift note. One functional of the force history, the part of the motion that Newton's refinement removes, is converted by $\hbar$ into a phase and by the mark cost into a statistical distance; for a constant force it is the chord's segment times $F/2v$.

7. Both factors are in the *Opticks*

Newton's optics supplies, on this paper's reading, the two spreads whose product is κ, in the form of two quantities of the probe; the identifications below are readings, and each says what it rests on.

A least length, measured. Book II Part III defines the interval of the fits as "the space it passes between every return and the next return" of the ray's disposition to be reflected, and Proposition XVIII gives it: for the confine of yellow and orange passing perpendicularly into air, "the Intervals of their Fits of easy Reflexion are the 1/89000th part of an Inch", that is about $0.285\,\mu$m. A datum obtained through the fits repeats with that period, so it locates the corpuscle no better; Newton's rings are a gauge of exactly this kind. We read it as $\Delta\hat{Q}_A$. The reading is an analogy: the interval is a period along the ray, a longitudinal length, while $\Delta\hat{Q}_A$ is the spread of a transverse mark's pointer.

An impulse, inferred. Query 29 asks whether the rays of light are "very small Bodies emitted from shining Substances". A body carries an impulse, and we read its scale as $\Delta\hat{P}_A$; Query 29 itself speaks of sizes, and it is a query, so the impulse scale is our inference. Newton has no way to measure it.

Colour dependence, measured. Observation 13 of Book II Part I reports the ratio of the red to the violet interval as "greater than as 3 to 2, and less than as 13 to 8", and "By the most of my Observations it was as 14 to 9". Query 29 makes the violet corpuscles the least and the red the biggest. The measured intervals and hypothesized sizes do not determine a momentum spread or the colour dependence of their product. A relation between size and momentum, a uniform lower bound and cross-colour constancy would each need an additional premise; none follows from these passages.

Inflexion. Query 1 asks whether bodies "act upon Light at a distance, and by their action bend its Rays", most strongly at the least distance, which is the effect that limits a mark's resolution when the beam is narrowed to improve it.

The interval times the momentum is a refraction invariant. Book II Part III Prop. XVII makes the intervals in two mediums "as the Sine of Incidence to the Sine of Refraction", from Observation 10 of Part I, and Prop. X states its result on the hypothesis that light is "swifter in Bodies than in Vacuo, in the proportion of the Sines", which Newton holds categorically in the *Principia*, Book I Section XIV. Together they make Λv, and so Λp, the same in every medium for each colour; that Newton's interval varies inversely with the corpuscle's speed, as de Broglie's wavelength does, is an old observation (Sakkopoulos, *Eur. J. Phys.* **9**, 123, 1988, metadata). A fit counted per interval along the path therefore counts the Maupertuis action $\int p \cdot dq$ in units of Λp, and the inscribed polygon and the parabola of Theorem 7 differ by $(F/v)\sum_j A_j/(2\Lambda p)$ fits, with A_j Lemma XI's tangent areas. Since the fit at arrival decides reflection or transmission (Prop. XII), the two are optically different corpuscles on Newton's own terms once that count reaches half an interval: on this reading his determinate fit would tell them apart on a single corpuscle, where the quantum phase is relative between two histories and shows only in a superposition. The two are different in kind. Newton had no superposition of amplitudes: imaginary quantities enter his algebra only as "impossible" roots, and his mechanism for the fits, in Query 29, is that the rays "stir up Vibrations in what they act upon, which

Vibrations being swifter than the Rays, overtake them successively, and agitate them so as by turns to increase and decrease their Velocities, and thereby put them into those Fits", a wave that disposes the corpuscle, with no adding of amplitudes. Addition of amplitudes arrives with Young's principle of interference (*Phil. Trans.* **92**, 12, 1802). Counting 2π per interval, a full return of the disposition, the fits phase is $4\theta_N$ at $\Lambda = \lambda/2$; the wave's own phase, which advances π per interval, differs by $2\theta_N$. Extending Props. X and XVII to a continuously varying speed is ours, and Prop. XV's oblique rule is more complicated than a count along the path.

These passages, with Prop. XI below, are held verbatim with line anchors in the source companion, from the fourth edition of 1730.

8. The junction and its additional physical premises

The modern results in §§3–6 assume quantum kinematics with supplied physical $\hbar > 0$. A position-coupled probe has conjugate reading-error and recoil operators, giving their standard-deviation product at least $\hbar/2$. For non-adaptive Gaussian probes, tests invariant under unknown initial position and velocity obey the stated action threshold with an explicit error–recoil correlation bound. The general disturbance theorem uses a stronger composite test for every pair of initial body states and charges spreads over its conditional and translated hybrid states. A later estimate of a kick does not erase that charged operator variance; conversely the variance theorem does not establish a universal posterior prohibition on classical kick inference.

The historical marking premise M3 proposed $\delta\Delta \geq \Lambda p$, including after later records. It is an analogy with the modern probe theorem, not an equivalent statement: hard interval widths and standard deviations differ, the interval of fits is longitudinal, and Query 29 does not supply a transverse momentum spread. The substitutions $\Lambda = \lambda/2$, $p = h/\lambda$ give $\Lambda p = h/2 = \pi\hbar$, rather than $\hbar/2$, and use modern quantum relations. The classical independent-box theorem retains its own availability and support assumptions in the canonical treatment.

Proposition XII describes a ray's returning disposition, and Query 29 proposes a corpuscular mechanism. These printed statements establish neither M3 nor a theorem negating every possible operational restriction. A determinate model **with additional record access** does furnish an exact countertest: allow the outgoing probe to travel freely to a screen at arbitrarily large distance D, keep its incoming momentum known and infer its outgoing direction from two position records. Recoil is then confined to width at most $p(\delta + \delta_s)/D$, tending to zero. This construction makes the required apparatus and limiting order explicit. With a remaining minimum position resolution, a fixed-force time threshold can persist; there is no uniform action floor across the shrinking-force family. The printed periodic disposition alone does not prove the required instrument availability or those limits.

The classical geometrical limit and quantum recording bounds therefore answer different questions. The former is consistent at arbitrarily fine mesh; the latter restrict resources in supplied quantum experiments. The source quantities motivate a counterfactual recording question, but their existence does not derive an irreducible cost, quantum states or universal calibration.

Theorem 7 uses another additional structure: a quantum central extension in which displacement and impulse compose up to phase $Jd/\hbar$. The refraction-invariant reading of Λp in §7 and the difference between counting pd before or after a kick motivate an action-counting analogy. That construction is the paper's interpretation. Newton's fits do not supply coherent superpositions, the Born rule or a unitary Weyl representation; cross-colour universality is only one of the additional requirements. The supplied scalar polygon phase is independent of the body's preparation, while observing it still requires a controlled coherent comparison.

A third comparison concerns Leibniz's continuity principle. In the stated unrestricted Gaussian invariant-mark family, the optimal error $\Phi(-\frac{1}{2}\sqrt{K_\tau/\kappa})$ is continuous at zero force exactly when $\kappa > 0$. Reading discernibility as the relevant outcome is a modern interpretive premise. Continuity of each fixed noisy apparatus is weaker than uniform continuity over a family with unrestricted repeats. A classical zero-recoil family with fixed total precision can also have a

uniformly continuous verdict. The continuity treatment and complete-record calculations retain those quantifiers. No general exclusion of classical completions follows from geometric or apparatus-by-apparatus continuity.

9. Newton and the classics

What Newton attempted

In the early 1690s Newton drafted a set of Classical Scholia to Propositions IV to IX of Book III, arguing that the oldest philosophers had known universal gravitation and its inverse-square law, so that the *Principia* recovered a lost wisdom rather than announcing a novelty. He never printed them. He did pass material to David Gregory, whose *Astronomiae physicae et geometricae elementa* of 1702 opens with the programmatic sentence that lest the physics delivered here seem something new and unheard of in astronomy, *eandem vetustissimis Philosophis notam ... ostendam*, I shall show it was known to the most ancient philosophers, naming Anaxagoras, Archelaus, Euripides and Pythagoras. That preface is the printed witness closest to the unpublished scholia, and the source companion holds its programmatic paragraph. The founding study is McGuire and Rattansi, "Newton and the 'Pipes of Pan' ", *Notes Rec. R. Soc.* **21** (1966), 108–143; the scholarship continues in Casini, "Newton: The Classical Scholia", *History of Science* **22** (1984), 1–46, and Mosley, "The Origins and Sources of Newton's Classical Scholia", *Erudition and the Republic of Letters* **9** (2024), 171–217; Schüller edited the scholia from Gregory's papers. All are verified at metadata level here and still to be read.

So Newton did draft a set of scholia founded on the classics, with Gregory's preface as their printed witness, and the attempt was doxographic. He sought ancient authority for a law he had already proved, and the authority he sought was for the *conclusions* of Book III.

Book I does carry two scholia that touch the ancient question, both from the geometers' side, and the first edition's Latin is held here (1687 witness; English renderings ours). The scholium after Lemma XI continues the series of contact angles without end: between any two, a new series "inseri potest", can be inserted, "Neq; novit natura limitem", and nature knows no limit. The scholium closing Section I opens by saying that the lemmas were put first "ut effugerem tædium deducendi perplexas demonstrationes, more veterum Geometrarum, ad absurdum", to escape the tedium of the ancient geometers' reductio: the method of exhaustion that Archimedes credits to Eudoxus, who used it to prove that the cone is a third of its cylinder, a result Archimedes says Democritus had asserted first without proof (preface to *The Method*). Newton prefers limits to the method of indivisibles, "quoniam durior est indivisibilium Hypothesis", answers the objection that ultimate ratios would imply ultimate magnitudes, so that quantity would consist of indivisibles, "contra quam Euclides de incommensurabilibus, in libro decimo Elementorum, demonstravit", and closes "cave intelligas quantitates magnitudine determinatas, sed cogita semper diminuendas sine limite": do not take them as quantities of determinate magnitude, but think of them as always diminishing without limit. Euclid is cited against least magnitudes, the ancients' exhaustion is replaced by limits, and the physical dispute about the cut, with the atomists who held least parts, is left aside. §6 reads that exhaustion through $\hbar$: for a figure traced by a motion, each round of it removes a phase.

The ancient arguments he passed over

The ancient material that bears on the *method* of Book I is the debate over division and the cut, and it is absent from the scholia. Six positions, each held here in a primary witness:

- **Democritus's cone** (Plutarch, *De communibus notitiis* 39). Cut a cone by a plane parallel to the base: are the surfaces of the two adjacent sections equal or unequal? If equal, the cone is a cylinder; if unequal, it is stepped. Chrysippus answers that the surfaces are

neither equal nor unequal while the bodies are unequal, and Plutarch calls that a licence to write whatever comes to mind.

- **Zeno's arrow** (Aristotle, *Physics* VI.9). At every now the flying arrow occupies a space equal to itself and so is at rest, and what is at rest at every now does not move. Aristotle's diagnosis is that the argument assumes time to be composed of nows.
- **The Mohist endpoint** (*Mozi*, Canons and Explanations). Halving terminates at an endpoint that cannot itself be halved.
- **Liu Hui's cutting** (commentary on the *Nine Chapters*, 263 CE). The finer the cutting the smaller the loss; cut and cut again until it cannot be cut, and the polygon coincides with the circle with nothing lost.
- **Hui Shi's stick** (*Zhuangzi* 33). A stick one foot long, halved every day, is not exhausted in ten thousand generations.
- **The leap and the time-atom** (al-Shahrastani on al-Nazzam; Maimonides, *Guide* I.73). The interval is divisible, yet it is crossed by leaps; and for the kalam, time is composed of atoms.
- **The Vaiśeṣika arrow** (*Vaiśeṣikasūtra* 5.1.16–18, layered, c. 100 BCE–200 CE, attribution to Kaṇāda traditional). The arrow's particular conjunctions are *ayugapat*, non-simultaneous, and that is the ground of the plurality of its motions; impulsion causes only the **first** motion, and each later one is caused by the *saṃskāra* deposited by the motion before it, until the impression is spent and gravity takes over.
- **Sound produced from sound** (*Vaiśeṣikasūtra* 2.2.36–37 in the Candrānanda recension, 2.2.31 in the vulgate). Sound arises from conjunction, from disjunction, and from sound, and is non-eternal, so what reaches the ear is a later member of a generated chain. Vātsyāyana's *Nyāyabhāṣya*, within the window at about 400 to 450 CE, argues the point from observation: the axe-blow is still heard at a distance after the axe-wood contact has ceased.
- **No motion at all** (Vasubandhu, *Abhidharmakośabhāṣya* ad IV.2, c. 350–450 CE). *na gatir yasmāt saṃskṛtaṃ kṣaṇikam*, there is no motion, because the conditioned is momentary; everything conditioned "is destroyed in the very place where it arose", so its passage to another place is impossible. Zeno concludes this and treats it as absurd; Vasubandhu concludes it and accepts it.
- **A medium that permits motion** (Umāsvāti, *Tattvārthasūtra* 5.17–18,

  c. 2nd–5th c. CE). *gatisthityupagrahau dharmādharmayor upakāraḥ*: the assisting of motion and of rest is the function of *dharma* and *adharma*. These substances push nothing; they are the standing condition without which motion could not occur, and space is given the separate office of accommodation.

- **Sound as spreading waves** (Diogenes Laertius VII.158, reporting Stoic doctrine). Hearing occurs when the air between is struck, "a vibration which spreads spherically and then forms waves and strikes upon the ears, just as the water in a reservoir forms wavy circles when a stone is thrown into it". The same water-ring image carries the Vaiśeṣika *vīcī-santāna*, so the model is attested independently in both traditions.
- **Continuity that fails below sense** (Epicurus, *Letter to Herodotus* 61–62, in Diogenes Laertius X). Atoms travel at equal speed through the void, and in aggregates they "move in different directions in times so short as to be appreciable only by the reason, but frequently collide until the continuity of their motion is appreciated by sense"; the assumption that continuity persists below observation "is not true in the case before us".
- **Least partless bodies** (Diodorus Cronus, reported at Sextus Empiricus, *Pyrrhoneion Hypotyposeis* III.32). Listed in the doxography of material principles between the atoms of Democritus and Epicurus and the unjointed masses of Heraclides. Diodorus argued from those minima that "a thing never is moving, but it has moved", the Greek twin of Vasubandhu's *na gatiḥ* three centuries earlier; the formula is at *Adversus Mathematicos* X

and is cited here without being read.

- **The voice as spherical waves** (Vitruvius, *De architectura* V.3.6–7, c. 25 BCE). The voice is propelled "by an infinite number of circles similar to those generated in standing water when a stone is cast therein", but "whereas the circles in water only spread horizontally, the voice, on the contrary, extends vertically as well as horizontally", so the water-ring image is corrected into a spherical wave by argument from the disanalogy.
- **The sling, applied to an orbit** (Plutarch, *De facie in orbe lunae* 923C–D, c. 100 CE). "The moon is saved from falling by its very motion and the rapidity of its revolution, just as missiles placed in slings are kept from falling by being whirled around in a circle", with 923D adding that "each thing is governed by its natural motion unless it be diverted by something else".

Lemmas X and XI perform exactly the operation these positions dispute. The scholia cite the ancients for what Book III concludes and leave them silent on how Book I proceeds.

The last four entries differ in kind from the rest, and they matter most here. The Greek and Chinese items are paradoxes about division, which set a problem. The Indian entries answer it, and they answer it three different ways: the arrow's flight is many motions carried by an impression (Vaiśeṣika), or it is no motion at all because each thing perishes where it arose (Vasubandhu), or it is possible only because a medium stands ready to permit it (Umāsvāti). Each also carries a stopping rule for division: the Nyāya *paramāṇu* than which nothing is smaller, the directional-parts reductio by which a touched atom would have parts, and the Jain atom that has no space-points at all. The Vaiśeṣika sūtras in particular state a **positive theory of propagation**: motion is a succession of numerically distinct events, each carried to the next by a quantity the previous one deposits, and transmission through a medium is a chain in which what arrives is a later member than what was sent. That is the structure the theorems quantify. The aperture bound of §10 is protocol-universal because the signal is a phase-space path whose total variation is unchanged by subdivision, which is the modern form of the claim that impulsion supplies only the first motion while the impression carries the rest. An ancient author had therefore already framed propagation as a bounded succession rather than a continuous traversal, and Newton's classical preface reached for none of it.

The hierarchy the ancient premises generate

Read as premises about records rather than about geometry, the cone and the arrow are the first members of a sequence whose next member is Newton's comparison. The apparatus of §3 covers all of them; what changes is the order of the signal against the nuisance it must beat.

Theorem 9. In the model of §3, with every mark obeying $\delta_j \Delta_j \geq \kappa$:

(i) *The arrow.* Distinguishing rest from uniform motion at speed v, with the initial position unknown, requires

$$m\, v^2 \tau \;\geq\; 8\, z_{1-\epsilon}^2\, \kappa, \qquad \text{that is} \qquad \tau\, E_{\text{kin}} \;\geq\; 4\, z_{1-\epsilon}^2\, \kappa.$$

(ii) *Newton.* Distinguishing uniform motion from constant force, with the initial position and velocity unknown, requires $F^2\tau^3/m \geq 96 z_{1-\epsilon}^2 \kappa$, which is Theorem 2.

(iii) *The cone.* Measuring a static taper requires nothing. With no recoil to propagate, the deflection grows without bound as marks accumulate, so a static solid has no floor.

Proof. For (i) the nuisance is the constant, so u ranges over vectors with $\sum_i u_i = 0$ alone. With $P(t) = vt$ the identity of §3 reads $u^{\mathsf{T}}P = v\sum_i u_i t_i = v\,T(0)$, and $T(\tau) = 0$ with Cauchy–Schwarz gives $|T(0)| = |\int_0^\tau T'| \le \sqrt{\tau}\sqrt{E}$. The noise bound $u^{\mathsf{T}}\Sigma u \ge 2\kappa E/m$ needs only $\sum_i u_i = 0$ and so is unchanged. Hence $d^2 \le v^2\tau E\, m/(2\kappa E) = mv^2\tau/(2\kappa)$, and $d \ge 2z_{1-\epsilon}$ gives the claim. For (ii) the nuisance is the linear space, $u^{\mathsf{T}}P = \frac{F}{m}\int_0^\tau T$, and the proof of Theorem 2 applies. For (iii) the marks leave the object unchanged, so $\Sigma = \mathrm{diag}(\delta_j^2)$; with N marks of resolution δ at each of two heights separated by Δh, the taper ϑ gives $d^2 = N\vartheta^2\Delta h^2/(2\delta^2)$, unbounded in N. □

The arrow's rung is attained by two marks: weights $(-1, 1)$ at 0 and τ make T linear, so the Cauchy–Schwarz step is an equality, with the first mark balanced, $\delta_0 = \Delta_0\tau/m$, and the last sharp. Newton's rung needs the dense limit of Theorem 2.

The two dynamical cases share one invariant. If the signal is the n-th order departure P with $P(0) = \cdots = P^{(n-1)}(0) = 0$, the quantity the theorem bounds below is

$$m\left(P^{(n)}\right)^2\tau^{2n-1} \;\gtrsim\; \kappa,$$

which has the dimensions of action for every n: at $n = 1$ it is $mv^2\tau$, twice the kinetic energy times the duration, and at $n = 2$ it is $F^2\tau^3/m$, twice $\tau\Delta E$. The theorem covers $n = 1$ and $n = 2$; for higher n the form is dimensional analysis. Newton's comparison is the second rung of a ladder whose first rung is Zeno's.

Why the arrow and not the sling

One feature of the list needs saying. Almost every ancient entry is rectilinear or static, while the case Newton uses to **define** centripetal force is the sling: Definition V has the stone whirled about in a sling endeavouring to recede from the hand, and names the force that "retains it in its orbit" after the planets "perpetually drawn aside from the rectilinear motions, which otherwise they would pursue".

The exception is Plutarch, and it is a pointed one. *De facie* 923C has the moon kept from falling "just as missiles placed in slings are kept from falling by being whirled around in a circle", which is Newton's apparatus applied to Newton's case a millennium and a half earlier. The inference runs the other way: for Plutarch the whirling *prevents* the fall, so speed sustains; for Newton the cord *pulls the stone inward*, and what it diverts is a straight line that would need no cause. The same object supports opposite accounts, and what separates them is which motion is taken to be free. Plutarch's next clause, that "each thing is governed by its natural motion unless it be diverted by something else", is as near as the passage comes, with natural motion still doing the work inertia later does. The Classical Scholia are reported to cite Plutarch's *De facie* among the ancients said to have known the doctrine of gravitation (Casini 1984; McGuire and Rattansi 1966; metadata, not read here), which would put the sling in a text Newton was mining for authority while writing the propositions it models; whether they cite this passage is open.

Theorem 9 explains why the exception stayed isolated, and Aristotle supplies the other half of the reason. *De caelo* I.2 makes uniform circular motion the natural motion of the aether, needing no cause, which is the exact inverse of Definition V. For the heavens, then, curved motion was the default and straight motion the anomaly, so the sling illustrated a conclusion already held. The claim that the ancients treat rest as natural holds below the moon and is incomplete above it; the reference is at metadata level here. The ancient question is the first rung, whether the thing moves at all, with position as the only nuisance. The sling is the second rung, whether a force acts, with uniform motion as a further nuisance, and that rung cannot be stated until straight-line motion is held to need no account. Ancient physics largely does not hold that, so the sling has no work to do in it. The Vaiśeṣika comes closest by charging only the first motion to impulsion, and stops short of making a straight continuation free and a curved one costly.

Indian sources do supply one whirled object, and use it for a third purpose again. The firebrand circle, *alātacakra*, appears in Vasubandhu (GRETIL source companion, Pradhan 189.23–24) to argue that because contact with the parts is successive, the awareness of a whole is really of the parts, and at Pradhan 33.9 with *āśuvṛttyā*, by rapid action. That is an argument about sampling, and it is the ancient form of what Theorem 9 says about records: within a window below the mesh a succession is indistinguishable from the continuous thing it mimics. The historical qualification for Plutarch's exception is in the source companion; the fuller arrow-and-sling discussion remains in the archived manuscript.

What each ancient premise buys

- **Zeno.** His conclusion holds of local records: marks confined to a window shorter than $\tau_{\text{arrow}} = 8z^2\kappa/(mv^2)$, used by themselves, are consistent with rest. Aristotle's diagnosis explains why, since a now carries no record and every datum is a window.
- **The Vaiśeṣika arrow and the chain of sounds.** These supply the positive half the Greek material lacks. Theorem 9's ladder can be read as their statement made quantitative, a reading, since the theorem prices marks and is silent on whether motion is a succession: each step carries a bounded quantity to the next, and reading any one step costs κ. The sound chain is the case where an ancient author asserts a finite propagation time and argues it from an experiment. Newton held that light takes time too, "about seven or eight Minutes of an Hour in passing from the Sun to the Earth" (*Opticks* II.iii Prop. XI); what his corpuscle lacks is the chain, since the same body arrives carrying its own disposition, where the Vaiśeṣika sound is a later member of a generated series.
- **Vasubandhu.** His conclusion is what the theorem says about local *records* rather than about the world. Within a window below the mesh the marks are consistent with rest, so no recorded instant exhibits motion and travel is never displayed, only inferred from enough instants together. The theorem declines his further step from the records to the world, and Theorem 9(i) measures exactly how many instants are needed.
- **Epicurus.** His is the closest ancient statement to what the theorems say about records. Motion presents itself as continuous to sense and is a succession below it, and he names the fallacy of extrapolating the observed continuity downward. He draws the boundary where sense fails; Theorem 9 draws it at a mesh computed from the mark cost, which is the difference between a threshold that is reported and one that is derived.
- **Umāsvāti.** The Jain medium is the one ancient category with no counterpart in the theorem, and its absence is informative. Theorems 2 and 9 need no medium: what they price is the mark, not the motion. A doctrine on which motion requires a permitting substance predicts nothing about the cost of observing it, which is the respect in which this entry sets a limit on how far the reconstruction reaches.
- **Democritus and Chrysippus.** Democritus's sections gave the first result the method of exhaustion was built to prove, the cone as a third of its cylinder, which Archimedes credits to him as asserted without proof and to Eudoxus as proved; the Galileo area is the same third, $A = \frac{1}{3}v\tau s$. The dilemma about adjacent sections dissolves with resolution alone, and case (iii) shows it needs no action floor. Adjacent recorded sections are equal, and the inequality of the bodies is recovered cumulatively, which is Chrysippus's answer given a definite sense. This corrects a heuristic recorded earlier in this programme, that the cone is the arrow rotated into Euclidean time: the static problem lacks back-action, and back-action is the entire source of the floor.
- **Newton's own objection.** Answered on his terms. The mesh τ_* is a least *recorded* duration and implies no least magnitude: it moves with F, m and the confidence demanded, so Euclid's incommensurables survive and the quantities of the geometry still diminish without limit, as the Section I scholium requires. What is bounded is the record, at the rate $\hbar$.

- **The Mohists and Liu Hui.** They are right about local records. A window shorter than the mesh $\tau_* = (96z^2m\kappa/F^2)^{1/3}$ cannot exhibit the force by itself, which is Corollary 3; cutting itself goes on, and the sharp protocol of Theorem 2 needs marks far denser than τ_*. What stops is local exhibition, not division, so the Mohist unhalvable endpoint is a reading the theorem does not give.
- **Hui Shi.** He is right about geometry, which is Proposition 1: the halving never ends, and nothing in the figure resists it.
- **Al-Nazzam.** The local record below the mesh is his leap. The interval stays divisible, and the crossing is exhibited only in finitely many steps.
- **Galileo, Cavalieri and Guldin (1621–1647).** The Newton-age dispute over indivisibles, the bowl and cone, the wheel's unquantified voids and Cavalieri's aggregates, concerned static figures. Case (iii) puts no floor there, and the scholium's limit doctrine is the right answer; the entries are in the continuity note, §4.
- **Leibniz (1687, 1704).** His law of continuity, stated in the year of the *Principia* as a test of laws of motion, holds for Newton's comparison exactly when $\kappa > 0$ if outcomes are measured by discernibility, his own "à peine distingué", and on both branches if they are measured geometrically; his *petites perceptions* are the additive shares of the cut measure (continuity note, §§1–3).
- **Newton's Rule III (1713).** Book III separates division by reason, "ex mathematica certum", from division "per vires naturae", "incertum", to be settled by one experiment, so Newton's stated physics leaves the physical limit open (continuity note, §4b).
- **The kalam time-atom.** Denied. The mesh depends on F, on m and on the confidence demanded, so it is a dynamical resolution rather than a universal atom of time. A floor on action coexists with a continuum of instants, and Theorem 9 shows the mesh moving as the force changes.

The section's claim about Newton is therefore double. He looked to the ancients for authority and found the wrong ancients, taking the doxography of gravitation while passing over the dispute about division that his own Lemmas turn on. And the premise that would have converted that dispute into a theorem, an indeterminacy in the least parts of light, is one the atomists approached through least parts, which the theorem replaces by a least action for records and no least magnitude, and one that what he printed about the fits excludes, in the single place where he had a measurement.

10. Unmarked preparations and prior art

Preparations of unbounded extent. Theorems 2, 5 and 6 concern protocols that mark the trajectory. A protocol that prepares once, waits and measures once faces a different bound. In that case the two hypotheses differ by the phase-space displacement $(-Ft^2/2m,\ Ft)$, whose components have product exactly $\tau\Delta E$, and Mandelstam–Tamm in the geometric form of Anandan and Aharonov (PRL **65**, 1697, 1990), with Helstrom's discrimination bound (*Quantum Detection and Estimation Theory*, 1976), gives

$$\frac{F\tau L}{\hbar} + \frac{F\tau^2 P}{2m\hbar} \;\geq\; \theta, \qquad \tau\Delta E \;\geq\; \frac{2m\theta^2\hbar^2\tau}{(2mL+\tau P)^2}, \qquad \theta = \arcsin(1-2\epsilon),$$

for an apparatus whose position and momentum spreads are bounded by L and P. It is the integrated form of a kick bound, $\int_0^\tau |f(t)|L(t)\,dt \geq \hbar\theta$ with $L(t)$ the body's position spread at time t: a force is a succession of impulses, and an impulse is registered only through the position spread it acts on (path-length note, Theorem K). The floor equals $\theta^2\hbar^2/(4LP)$ at the balanced aperture $L = \tau P/2m$ and falls to zero as either side grows at fixed area, so a squeezed state of area $\hbar/2$ evades it as a large one does. In Newton's quantities the first inequality is $F\tau \cdot L + \frac{F\tau^2}{2m} \cdot P \geq \hbar\theta$, the impulse of Proposition I against the position aperture plus the sagitta

of Lemma X against the momentum aperture. It holds for every finite adaptive protocol of instruments, of any Kraus rank and with any apparatus memory, because the displacement is a phase-space path whose total variation does not grow when it is subdivided. The bound is tight: two narrow packets separated by $\pi\hbar/(F\tau)$ decide the comparison with error probability of order $\tau\Delta E/\hbar$, for arbitrarily small $\tau\Delta E$, at the price of an apparatus whose extent diverges. So the floor of order $\hbar$ requires either that the trajectory be marked or that the laboratory be bounded, and a non-Gaussian preparation of unbounded extent evades it.

One accounting. The kick bound pays for the force at the body and Theorem 6 pays for it at the marks, and any split of the force between the two gives a valid bound. For a force of one sign and recoil-only marks, provided the body and recoil spreads have common bounds uniform over the entire split family and its hybrid states, minimization of this upper cost is pointwise, $\hbar\theta \le \int_0^\tau |f(u)| \min\big(L(u), R(u)\big) du$, where $R(u) = \frac{1}{m}\sum_j \Delta_j G_\tau(t_j, u)$ is the recoil length, the marks' recoil spreads weighted by the displacement each produces at time u relative to the chord and summed, with G_τ the Dirichlet Green's function of the interval: at every moment the force is paid in the smaller of the two lengths (path-length note, Theorem M).

Prior art. The combination $F^2\tau^3 \gtrsim m\hbar$ is the standard quantum limit for detecting a force on a free mass. It goes back to Braginsky and Vorontsov (*Sov. Phys. Usp.* **17**, 644, 1975); the canonical statement is Caves, Thorne, Drever, Sandberg and Zimmermann, *Rev. Mod. Phys.* **52**, 341 (1980); the standard accounts are Braginsky and Khalili, *Quantum Measurement* (1992), and Bocko and Onofrio, *Rev. Mod. Phys.* **68**, 755 (1996). Its status has been disputed since Yuen's objection: Caves defended it (PRL **54**, 2465, 1985) and Ozawa exhibited a measurement breaking it for free-mass position (PRL **60**, 385, 1988). The dense-mark limit of §3 is the continuous position measurement of Caves and Milburn (PRA **36**, 5543, 1987), with the force limit worked out in Clerk, Devoret, Girvin, Marquardt and Schoelkopf (*Rev. Mod. Phys.* **82**, 1155, 2010); what Theorem 2 adds to that model is the invariant-test criterion at a fixed error probability and the sharp constant. Hypothesis testing of a force waveform on a quantum probe is Tsang and Nair (PRA **86**, 042115, 2012), the closest prior art to §10's framing, and a quantum-mechanics-free subsystem removes the back-action floor altogether (Tsang and Caves, PRX **2**, 031016, 2012). Theorem 4's coupling is von Neumann's (1932), its inequality Robertson's (1929) and its ancestor the Heisenberg microscope (1927); Bohr and Rosenfeld (1933) and Araki and Yanase (1960) set limits of this kind from field measurability and from conservation laws. That preparations beat the limit is Yuen's point (PRL **51**, 719, 1983); the two-packet preparation of this section is the delocalized force sensor of Childs, Preskill and Renes (*J. Mod. Opt.* **47**, 155, 2000), and its sensitivity to displacements below the uncertainty area is that of Zurek's sub-Planck structures (*Nature* **412**, 712, 2001; Toscano, Dalvit, Davidovich and Zurek, PRA **73**, 023803, 2006). The correlated probes of §4 apply the mechanism of Yuen's contractive states to the meter, as Braginsky and Khalili's back-action-evading measurements do, and the grid probes of §5 are the grid states of Gottesman, Kitaev and Preskill (PRA **64**, 012310, 2001). The ingredients of §§5–6 and §10 are standard: Ozawa's disturbance operators, Mandelstam–Tamm in the Bures angle (Uhlmann 1992; Anandan and Aharonov 1990), hybrid arguments (Bennett, Bernstein, Brassard and Vazirani 1997), Bures-angle speed limits for processes (Taddei et al. 2013), the position-spread form of force sensitivity (Tsang, Wiseman and Caves 2011), symmetry-based information–disturbance bounds of Wigner–Araki–Yanase type, and the Strang splitting behind Theorem 7. All these are cited at metadata level. This paper collects the quantified protocol and instrument bounds, their explicit constants, the split accounting of §10, the phase identities of §6 read in Newton's and Archimedes' areas, and the stated physical premises. Literature priority for that combination has not been established. The inequality $F^2\tau^3 \gtrsim m\hbar$ itself is standard.

The classical theorem belongs to a literature it does not cite. Worst-case recovery of a linear functional from noisy linear data is optimal recovery in the sense of Micchelli and Rivlin (1977), and the apparatus series this programme built, together with the worst-case form of §3,

are results of that kind; information-based complexity (Traub, Wasilkowski and Woźniakowski, 1988) is the same setting. The invariant-test step of Theorem 2 is standard (Lehmann and Romano). A referee from either community will ask why these go unmentioned, and the honest answer is that the theorems were derived without them.

10a. Kepler completion and the action-selection premise

Dated revision, 2026-10-02. The marked-comparison theorems assume a positive action scale. Here $\hbar$ denotes the reduced phase constant, written h in the self-sufficiency and reachability notes; the full Planck constant in this paper remains $h = 2\pi\hbar$. To investigate whether motion itself requires a positive scale, the maintained self-sufficiency proof compares classical and quantum point-Coulomb dynamics. Classical radial inward or bound data collide in finite time and have no continuation under the original force field. The non-relativistic Coulomb Hamiltonian is self-adjoint on $H^2(\mathbb{R}^3)$ for every $\hbar > 0$, and its evolution exists for all times. That comparison establishes an incomplete classical flow and one global completion. It does not exclude classical regularizations or other completions.

The same proof now establishes the classical comparison on a fixed collision-free interval. For a fixed orbit with minimum radius $\rho_{\min} > 0$, fix $0 < \delta < \rho_{\min}$ and a real $V_\delta \in C_b^\infty$ equal to $-k/r$ outside $r < \delta$. For the normalized coherent preparations with width $\ell_\hbar = \sqrt{\hbar L_*/P_*}$, fixed $L_*, P_* > 0$, centered weighted Sobolev estimates and Hardy's inequality give

$$\sup_{0\le t\le T} \|(V - V_\delta)U_\hbar^{(\delta)}(t)\varphi_\hbar\|_2 = O(\hbar^{3/2}) = o(\hbar).$$

Duhamel transfers the smooth-potential Gaussian approximation to the Coulomb evolution. Separate energy and position estimates give $\langle x\rangle = q_t + O(\sqrt{\hbar})$ and $\langle p\rangle = p_t + O(\sqrt{\hbar})$, uniformly on the fixed interval. The constants depend on T, the preparation, derivatives of V_δ and $(\rho_{\min} - \delta)/2$. No collision, long-time or shrinking-δ limit follows. Hepp's theorem supplies the smooth-potential limit [@Hepp1974; passage, §2]; the singular-multiplier and moment estimates are the October 2 written derivation in §5a of the maintained proof, which has not received independent verification.

The packet limit and the complete record of a coherent swarm test different preparations and observables. The reachability proof, Theorem 5, shows exactly for quadratic forces that an incoherent mixture of shrinking coherent packets recovers Newton's branch-weighted coordinate density at each fixed time and regular record value. Its coherent-stream contrast instead requires coherence across launch points and a record that resolves the interference. The same classical swarm does not choose between those attachments. A physical rule that selects and preserves the required coherence remains an additional obligation.

At finite c, Proposition 2 of the self-sufficiency proof distinguishes the selection premises. For Dirac–Coulomb, essential self-adjointness gives $\hbar \in [2k/(\sqrt{3}c), \infty)$; requiring finite separate potential energy gives $(k/c, \infty)$; a critical Schur-domain or the specified cutoff limit extends the latter to $[k/c, \infty)$ [@EstebanLewinSere2019; passage, Theorems 1.1 and 1.11, arXiv v7]. The critical domain is defined by the original Esteban–Loss closed Schur form [@EstebanLoss2007; passage, Definition 3 and Theorem 4]. The maintained proof records an angular-channel counterexample to the literal compact domain formula printed in arXiv v7, equation (1.32); that rewrite is not a proof premise. If extra boundary conditions are allowed, self-adjoint Dirac realizations exist at every $\hbar > 0$. For Klein–Gordon, strict coercivity in the usual energy topology and the cited charge-space construction give $(2k/c, \infty)$ [@LangerNajmanTretter2008; passage, Example 8.3, Theorems 8.5–8.6]. Its principal radial Friedrichs operator includes $\hbar = 2k/c$ with a no-log branch [@DerezinskiRichard2017; passage, §2.3 and Theorem 5.5]. That radial selection is not a proof of full critical Klein–Gordon dynamics: the critical combined form does not control the singular multiplier needed in that construction.

Probability conservation justifies Dirac self-adjointness; it does not choose an extension. Finite separate energies, a particular point-source limit, Friedrichs closure and an energy topology are further premises. Before a selection-dependent threshold can establish action necessity, those premises must be justified independently of the desired gap. There is consequently no single unqualified admissible interval $[\hbar_{\min}, \infty)$ here, and no continuum Yang–Mills mass-gap result follows from this mechanical comparison. The action threshold concerns a family of selected theories; a spectral gap concerns a fixed operator.

11. Obligations before submission

The historical obligations and the scope of the new mathematical material are stated separately so that a reader can see what the present draft rests on.

1. **Historiography, to be read rather than cited.** On the *Opticks* side, Shapiro's *Fits, Passions, and Paroxysms* (1993) on the theory of fits and how determinate Newton meant it, which is where §8's central claim must be tested, and Sabra's *Theories of Light from Descartes to Newton* (1981) for the context of Query 29. On the *Principia* side, Guicciardini's *Reading the Principia* (1999) and his *Isaac Newton on Mathematical Certainty and Method* (2009) for what the limit arguments of §2 were for, De Gandt's *Force and Geometry* (1995) directly on the sagitta, and Bertoloni Meli's *Thinking with Objects* (2006) on the sling and the pendulum as objects to think with. On the scholia, McGuire and Rattansi, "Newton and the 'Pipes of Pan' " (1966), beside Casini and Mosley. On the ancient comparison, Sorabji's *Time, Creation and the Continuum* (1983), which treats Zeno, Diodorus Cronus, Epicurean minima and kalām atoms together; von Rospatt (1995) on momentariness; Dhanani (1994) on kalām atomism and the Indian influence question; and Lloyd and Sivin's *The Way and the Word*

   (2002) as the methodological standard that protects §9 from the charge of naive parallelism. Clagett (1959) for impetus, where Philoponus' *rhopē* is the Western counterpart of *saṃskāra*.

2. **Editions.** The *Opticks* passages come from a transcription of the 1730 fourth edition and should be cited from that printing; the *Principia* passages should be cited from Koyré and Cohen's variorum for the Latin, the first edition of which is now held for Book I Section I, and from Cohen and Whitman for the English, in place of the Motte text and this paper's own renderings. The firebrand passages of the *Abhidharmakośabhāṣya* are to be added to that companion, and whether the Classical Scholia cite *De facie* 923 itself is to be checked in McGuire and Rattansi and in Casini.
3. **October 2 mathematical additions.** The localization, moment estimates and replacement selection proposition are written derivations supported by the cited source passages. The September 23 and October 1 reviews predate them and supply no independent verification of these additions. Their stated domains and fixed-data limits must be retained in submission. Any stronger action-necessity claim requires an independent physical justification of the selection premise; full critical Klein–Gordon dynamics remains open.

Three mathematical items remain open and are not obligations of the paper: the worst-case interval $9 \le F^2\tau_*^3/(m\kappa) \le 36$ of §3, which concerns a framework without quantum instances; whether the minimum of the one accounting of §10 over splits and lines is attained by some protocol, which would make it the exact floor; and whether the constants $1/8$ and $1/2$ of (5) are attained, the line of best uniform approximation being optimal for the momentum term alone.

12. Consequence for STATE

This synthesis collects the conditional mark bounds, disturbance estimates and polygon phases together with Newton's source materials. The quantum results supply a positive canonical scale; their record costs do not derive that scale from Newton's axioms. The foundations content is §§3–6: the sharp Gaussian floor, the mark cost, the disturbance floor for every instrument and the phase that no preparation of the body moves. The history content is §§7–9, resting on the source companions. Section 8 states the limits of their junction: the fits period and inferred corpuscular momentum are not measured transverse uncertainty spreads; deterministic remote-readout countermodels require explicit instrument access; and the action-counting analogy supplies neither quantum composition nor a universal scale. The conditional refraction invariant and phase comparison retain their stated interpretive premises. Section 11 records this draft's historical obligations and the proof status of the mathematical additions. The October 2 §10a separates the closed fixed-interval Kepler localization and first-moment problem from the open physical selection, coherence and action-necessity obligations. STATE maintains the research queue, including the continuum Yang–Mills obligations.